Influence of the musical excerpt envelope on the relative modal threshold

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Abstract - *This study examines the relationship between the temporal decay threshold for detecting modal resonances and the envelope characteristics of musical excerpts, focusing on low frequencies (below 250 Hz). Modal decays were simulated using a code-based model of a small, acoustically optimized room for critical listening (e.g., music production, mixing, or mastering). An ABX test was conducted using two musical excerpts: one containing real drums and the other featuring synthesized drums. Each excerpt was compared to its modified versions, which incorporated the effects of three modal decays selected based on thresholds established in prior research. The results indicate that the decay times previously thought to produce audible effects were not significantly detectable for real drums. Only the comparison between the original synthesized drums and those modified with the longest decay time showed a significantly detectable difference. These findings imply that the temporal properties of musical content may be crucial to perceiving modal resonances and are influenced by specific musical stimuli, such as instrument type. These findings highlight the necessity for further research on objective metrics that link musical temporal characteristics with modal resonance perception, providing insights for optimizing acoustic space design.*

1. INTRODUCTION

Modal resonances in small rooms (volumes less than 250 m^3) remain a significant concern for studio owners and acoustic designers. While the perception of acoustic phenomena such as single reflections and reverberation is well-documented, the perception of modal resonances, particularly in low-frequency content (below 250 Hz), remains less explored. Traditional reverberation analysis tools, which assume diffuse sound fields, are inadequate for analyzing the perceptual effects of low-frequency modal resonances in small rooms.

Although sound decay analysis for resonances and reverberation is similar, traditional tools such as octave band filters inadequately characterize low-frequency phenomena by averaging decay rates across adjacent frequencies. In contrast, narrow-band filters tend to modify the temporal aspects of the signal due to their own non-ideal temporal response. These are examples of why traditional tools are not particularly practical for low-frequency analysis.

Historically, studies and modeling of the modal phenomenon have focused on wave-based approaches, developing room design criteria that attempt to link objective metrics to perception and preference for good acoustic design, as seen in Bonello (1979) through the so-called Bonello criterion. Recent developments employed room acoustic simulations to identify optimal dimensions by iteratively varying room dimensions and estimating response quality (Cox et al., 2004).

Traditional methods for controlling acoustic reflections, while being effective for mid and high frequencies, are impractical for low frequencies due to the significant space they require. This limitation is increasingly common as available spaces become smaller.

Recent methods for modal control have included digital processing techniques, such as pole-zero filter design in Mäkivirta et al. (2003), which is similar to the REW software processing approach (Mulcahy, 2024). Other examples of state-of-the-art modal control include secondary radiator systems with digital processing for active absorption or acoustic impedance control (Boulandet, 2012).

Despite extensive research being conducted on objective modal phenomena, there remains a significant gap in understanding the perception and preference of these phenomena, particularly in relation to musical content. Understanding how different musical excerpts interact with room modes is essential for optimizing listening environments and ensuring accurate sound reproduction in music production. As Fazenda (2004) highlights, traditional studies often rely on single resonances and monotone signals, which fail to capture the complexity of real-world scenarios involving music and multiple resonances.

Recent studies (Fazenda et al., 2015) used modal decomposition methods to simulate room impulse responses and process audio to include simulated resonance effects. They conducted listening tests using these processed audios to establish the absolute temporal decay threshold for detecting resonances (absolute modal threshold) and relative thresholds specific to the

test stimuli category (artificial or natural). However, these studies did not include an objective signal analysis to correlate objective audio metrics with the results of relative thresholds.

The current study introduces a listening test in an ABX format to investigate whether resonance detection depends not only on the decay time of the room response but also on the temporal characteristics of the original signal itself. To achieve this, two bass drum audio samples were chosen, one from real drums and another from synthesized drums. These excerpts were digitally processed by adding three different modal decay times, determined based on the modal thresholds established byFazenda et al. (2015).

The hypothesis of this investigation is that the detectability of modal resonances depends on the temporal characteristics of the musical excerpt, with synthesized drums (characterized by shorter decay times) being more likely to reveal these effects than real drums, which exhibit longer and more irregular decay patterns.

2. METHODS

The simulation method was based on the modal decomposition model described by Kuttruff (2000). It employs Green's function, which solves the Helmholtz equation derived from the wave equation, representing a point source inside a cuboid-shaped room with rigid or semi-rigid boundaries. This approach yields the transfer function for pressure at a specific point within the room.

The pressure at position $\bar{r}$ due to a source at $\bar{r_0}$ is given by Equation (1).

$$p_{(w,\bar{r})} = \sum_{n_x}\sum_{n_y}\sum_{n_z} \frac{A_{w_n(\bar{r},\bar{r_0})}}{(w^2 - w_n^2 - j2w_n\delta_n w)} \quad (1)$$

where n is the order of the modes that are taken into account. x, y, and z refer to the length, width, and height directions of the simulated room. A_{w_n} represents the amplitude of the pressure related to the position of sources and listening point, w_n are all the modal frequencies, dependent on the dimensions of the room. w is the angular frequency and δ_n is the damping factor of each resonance.

The 60 dB decay time is related to the damping parameter, which controls the energy decay rates. This relationship follows Equation (2) as presented in Antsalo et al. (2001).

$$\delta_n = -\frac{1}{T_{60}} \ln 10^{-3} \approx \frac{6.908}{T_{60}} \quad (2)$$

Based on this theory and previous implementations of similar methods, a simulation was carried out through the development of a Python 3 code to generate impulse responses of a determined enclosure. The focus was on the decay times of interest for the test, denoted as T1, T2, and T3. These times were selected from threshold determinations by Fazenda et al. (2015), targeting the different time zones for the detection of resonances, from zone 1, where the effects are inaudible, to zone 4, where effects are clearly audible. Zones 2 and 3 represent the acceptability thresholds for artificial and natural stimuli, respectively.

The simulation operates with a frequency resolution of 0.125 Hz, applying desired decay times from 20 to 300 Hz, linearly decreasing to zero seconds up to 1 kHz. This approach serves three purposes: first, the aim is to represent a situation in a critical listening context, particularly in the low-frequency range. It is assumed that the rest of the spectrum is usually controlled or easier to control; secondly, the effects added by resonances outside of the low-frequency range are not of interest for this study; and thirdly, if some reverberation exists above 1 kHz, adding low-frequency absorption should not alter the room response above that part of the spectrum, meaning that the reverberation condition should remain consistent for the different simulated low-frequency decay times and the original excerpts.

Responses were generated up to 1 kHz and filtered with a 4th-order Butterworth low-pass filter at 500 Hz. The room dimensions were selected based on the ratios recommended by previous research (Fazenda et al., 2015) as optimal for the acoustic response and typical of modern studio sizes. The simulated dimensions of the room were X = 6.97 m, Y = 5.32 m, and Z = 2.69 m, with a volume of approximately 100 m^3. The source was at coordinates (0, 0, 0), and the listening point was at (3.16, 1.97, 1.3), in meters, as shown in Figure 1.

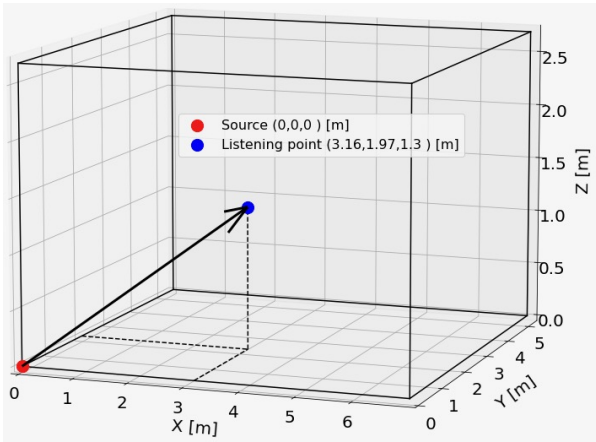

Figure 1: 3D sketch of the simulated room, with source and listener positions.

Resonances were modeled up to mode order 41; beyond this order, all resonances fall above 1 kHz, making higher orders irrelevant for the experiment's purposes. A reduction was applied to extract resonances falling below 1 kHz from the 68921 elements corresponding to all resonance frequencies calculated with order 41, resulting in a total of 11308 resonance frequencies.

The low-frequency responses were simulated using the previously described method, and then equalized in REW to flatten the responses below 250 Hz. Next, these were crossed at 250 Hz with a fourth-order Butterworth filter and combined with an ideal full-range impulse to obtain a response that is relatively flat across the entire

spectrum (±1.5 dB) with modal decays added only below 250 Hz. The 60 dB decay times curves calculated for these responses are illustrated in Figure 2, along with the threshold zones previously described.

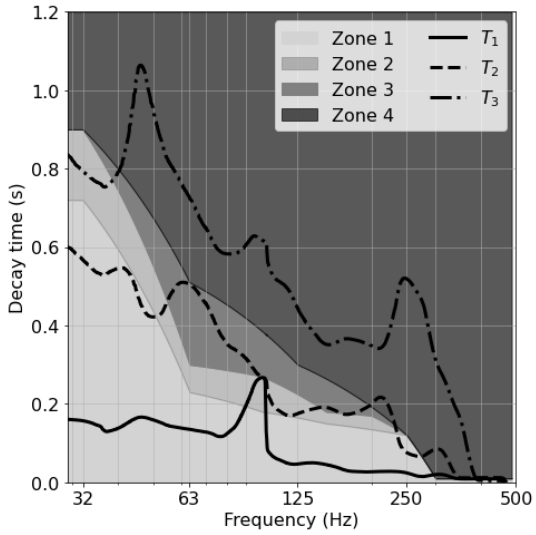

Figure 2: Decay time for the equalized responses.

The curve for T1 falls mainly in Zone 1, where times are below the absolute detection threshold. T2 is predominantly located between Zones 2 and 3, which are acceptable for artificial and natural stimuli, respectively. T3 falls almost entirely in Zone 4, where the sound effects are clearly audible according to previous studies.

After obtaining the impulse responses, base sound stimuli were selected for testing. One stimulus was a sample from the University of York (2019), categorized as "Synthesised Drums" ("SD"). This audio, consisting of three kick drum hits, two snare drum hits, and sporadic hi-hat strikes from a synthesized drum kit, lasted 2.331 seconds and could loop seamlessly, providing continuity. It featured rich low-frequency content and very short, controlled temporal decay.

The second stimulus was an audio excerpt of kick drum hits from a real drum kit, taken from the first bar of Fleetwood Mac (1977), lasting 3.160 seconds. The kick drum hits in this excerpt exhibited longer and irregular temporal decay, which potentially masked the effects of modal resonances, reducing the detection capacity. This excerpt was labeled "Real Drums" ("RD").

The process involved convolving the original excerpts with each of the three impulse responses obtained, adding the effects of resonances to the low-frequency musical content. Original and modified excerpts were normalized to integrated LUFS levels of -26 dB, according to the ITU-R BS.1770-5 standard (ITU, 2023).

The decays of the drum hits were characterized by using the Hilbert transform to obtain the amplitude envelope curve over time. A 4th-order Butterworth low-pass filter at 250 Hz was applied to focus on the frequencies of interest.

Figure 3 compares a kick drum hit from the "SD" audio in its original version and the altered version with the longest decay time (T3). Similarly, Figure 4 presents a comparison using the real drum sound of "RD". The original "RD" already exhibited prolonged resonances, making processing differences less discernible compared to "SD", where changes were more apparent, at least visually, over the envelope.

Notably, the original "SD" audio showed a linear decay, unlike the double-slope decay in "SD T3". In the real drum comparison in Figure 4, no obvious differences were observed between the original and the "RD T3" versions[1]. Based on this, "SD" was expected to better expose the resonances and thus allow subjects to more clearly detect them during the listening test than "RD".

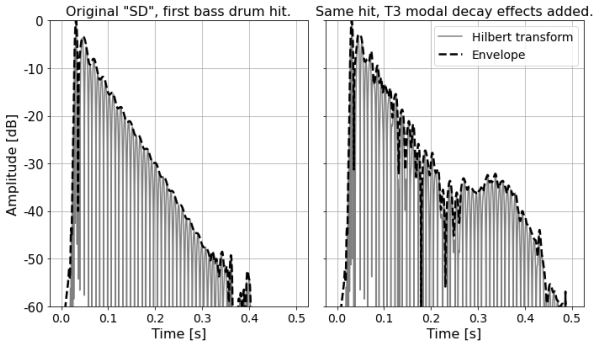

Figure 3: Comparison of envelope decay curves for Synthesized Drums (original vs. T3 modal decay).

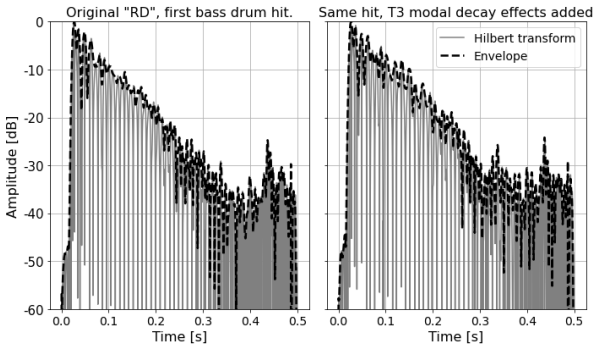

Figure 4: Comparison of envelope decay curves for Real Drums (original vs. T3 modal decay).

A double-blind ABX test was performed to detect differences between the original audio fragment and three versions with added decay times (T1 $<$ T2 $<$ T3). The test compared the original "SD" and "RD" audio with their modified versions. Each participant completed a total of 24 responses. These responses consisted of comparisons between the original and each modified version, as well as the reverse order of these comparisons.

This bidirectional approach allowed for situations where A was the original and B was modified, and vice versa, to account for potential order effects. For each excerpt, all possible combinations of original versus

[1]This analysis is preliminary, and consideration is being given to future developments on alternatives based on the autocorrelation function.

modified versions were tested, with each comparison repeated twice. This resulted in 4 responses per test for each pairing of original and modified versions.

Two versions of the test were designed with different randomized comparison orders, and participants were assigned to either version through a random link.

The ABX test was implemented using the open source platform "ABX Test", which presents a user guide published on GitHub by Pasanen (2021). The test was conducted via a web environment, designed using a ".yaml" code development. The graphic interface of the test is shown in Figure 5 where the first comparison of the test is presented, in this case showing a comparison based on "RD" where option B is selected to answer "X is B". X was randomly chosen by the test platform between A and B in each comparison.

Regarding the sound level settings for the test, participants were strongly advised to use headphones and adjust the listening level to their personal comfort, to ensure that they could fully focus their attention on the sound of the kick drum.

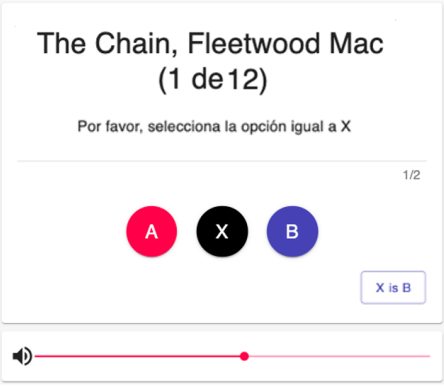

Figure 5: ABX app used for the online test.

Before the listening test, participants completed a questionnaire that focused on two main categories: personal experience with audio, sound, and music; and the quality of their audio equipment. It collected information on age, professional or amateur experience in sound and music, type and quality of headphones, and playback device specifications.

Exclusion criteria were established based on responses that indicated low-quality equipment or a lack of relevant personal experience in music or audio. This approach ensured that participants had sufficient expertise and high-quality equipment to effectively evaluate the audio samples in the study.

The results from the online tests were grouped, summarized, and subjected to statistical analysis using Python implementations for plotting, binomial calculations, and chi-square tests.

3. RESULTS

The study involved two participant groups: Group 1, consisting of four audio professionals (average age: 26 years), and Group 2, comprising two music-related professionals (ages 27 and 28). An additional audio professional was analyzed separately.

The listening test data were analyzed using the ABX methodology, employing binomial probability and chi-square tests. The null hypothesis assumed a 50 % success rate in identifying differences. The binomial distribution test was used to determine the probability that the participants correctly identified the differences, while the chi-square test assessed the statistical significance of these results.

For Group 1, three comparisons yielded a binomial probability of less than 0.05: RD vs. T1, RD vs. T3, and SD vs. T3. However, only the comparison involving the longest decay time (T3) with Synthesized Drums (SD) showed a binomial probability < 0.01 and a p-value < 0.05 in the chi-square test, indicating a statistically significant difference exclusively for this case.

Group 2 exhibited similar results, with only the SD vs. T3 comparison surpassing the 95% confidence threshold in the binomial distribution. This comparison produced a binomial probability < 0.01 and a p-value < 0.05 in the chi-square test.

When analyzing all participants, as shown in Figure 6, although the RD vs. T1 and RD vs. T3 comparisons surpassed the 95% confidence threshold in the binomial distribution, they did not achieve a significant p-value of less than 0.05. Significant differences were consistently observed for the SD vs. T3 comparison, with a binomial probability of less than 0.01 and a p-value < 0.01 in the chi-square test, as presented in Table 1.

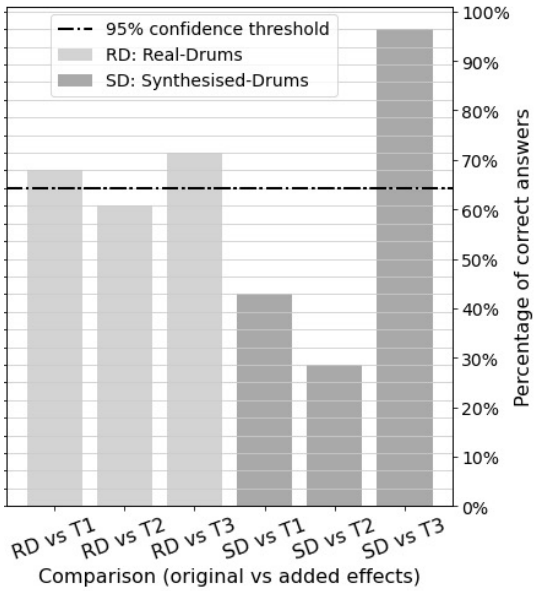

Figure 6: All test results summarized

Table 1: Statistical values for all tests summarized.

Comparison	Bin. Prob.	χ^2	P-value
RD vs T1	**0.03**	1.79	0.18
RD vs T2	0.08	0.64	0.42
RD vs T3	**0.01**	2.57	**0.11**
SD vs T1	0.11	0.28	0.59
SD vs T2	**0.01**	2.57	**0.11**
SD vs T3	**<0.01**	**12.00**	**<0.01**

These results suggest that differences between original and modified audio samples are consistently perceptible only when the longest decay time (T3) is applied to Synthesized Drums. Comparisons with shorter decay times (T1 and T2) and those involving Real Drums (RD) generally did not show statistically significant detectable differences.

A binomial probability of less than 0.01 indicates that the likelihood of the observed results occurring by random chance is below 1%. Similarly, a chi-squared goodness-of-fit test with a p-value below 0.01 suggests that the differences between the observed values and the expected values under the null hypothesis are statistically significant, with a confidence level exceeding 99%.

4. DISCUSSION

The results indicate that the detection of modal effects is highly influenced by the original characteristics of the musical excerpt, consistent with the findings of Fazenda et al. (2015). This alignment is particularly evident when comparing artificial stimuli, such as Synthesized Drums, to natural stimuli, such as Real Drums, as the added effects of modal decay in artificial stimuli are more easily detected.

These results have practical implications for optimizing critical listening environments, highlighting the need to tailor acoustic treatments based on the predominant musical content in each space. By aligning the acoustic design with the intended use of the room, it is possible to avoid unnecessary or overly expensive treatments in cases where such modifications may not significantly impact the listening experience.

However, this study reveals discrepancies with the literature, such as higher temporal values for detectability thresholds. Specifically, some decay times previously identified as easily noticeable were not significantly detected in the current test. This is particularly evident with real drums, where none of the comparisons showed statistically significant correct identification, despite the fact that the T3 decay time curve falls within the zone associated by Fazenda et al. (2015) with clearly audible effects.

These discrepancies may be due to the specific stimuli used, participant selection criteria, and less controlled listening conditions compared to previous research.

5. CONCLUSIONS

In conclusion, the results align with the study objectives, suggesting that the detection of modal effects depends on the specific characteristics of the musical excerpt. Statistical analysis indicates the following.

- In Group 1, three comparisons exceeded the 95% confidence threshold, but only one (SD vs. T3) exhibited a statistically significant p-value below 0.05, indicating detectable effects only in this specific case.
- Similarly, in Group 2, only one comparison (SD vs. T3) surpassed the 95% confidence threshold with $100\ \%$ correct responses and a significant p-value, reinforcing the detectability of the added effects only in this particular comparison.
- In all tests, this trend is consistently confirmed with a confidence level exceeding 99% in the goodness-of-fit test.

These findings suggest that modal detection is strongly dependent on the characteristics of the musical excerpt, highlighting the importance of developing objective metrics to characterize musical content when assessing perceptible differences related to room resonances. Future research should focus on quantifying the temporal characteristics of musical excerpts to improve the prediction of modal resonance detectability in different listening environments.

The current study is limited by its focus on two specific musical excerpts and a small pool of participants, which may affect the generalizability of the results. Future research could expand the range of stimuli by incorporating different instruments and musical genres to explore how the modal detection thresholds vary with stimulus type. Such efforts, along with the development of objective metrics for quantitative audio analysis, could inform the design of critical listening environments by emphasizing the importance of considering the temporal characteristics of musical content related to the intended uses of each space.

Additionally, enhancing the simulation model, exploring alternative methods, and implementing techniques to adjust simulated impulse responses directly in Python would provide greater control over simulations, reducing the dependence on external software for equalization.

Future evaluation of subjective preferences related to decay time could also be of interest, as preferences for reverberation time in mid and high frequencies are well-studied, but less so for modal resonances at low frequencies. Finally, future analysis of the relationship between the temporal perception of conventional reverberation (associated with the diffuse field) and low-frequency modal resonances could provide valuable insights into their perceptual differences.

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